

Parsa Oryani

M.Sc. Computer Engineering – Secure Computing

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Website | GitHub | LinkedIn

Research Profile

M.Sc. student in Secure Computing at Sharif University of Technology. Current research focuses on **security and privacy of LLM agents and multi-agent systems**, particularly information-flow control, authorization, provenance, secure memory/tool interactions, and runtime enforcement. Background in deep learning, cryptography, distributed systems, blockchain security, and formal security analysis.

Education

Sharif University of Technology Sep 2025 – Present
M.Sc. in Computer Engineering, Secure Computing Tehran, Iran

- National Master’s Entrance Examination Rank: 56. Research interests: LLM-agent security and privacy, information-flow control, authorization, provenance, and secure agent architectures.
- Selected coursework: Applied Cryptography (18.9/20), Deep Learning (18.35/20, Dr. Soleymani), Secure Software Systems (18.2/20), Formal Methods in Information Security, and Foundations and Applications of Blockchain.
- Deep Learning: Transformers, LLMs, state-space and generative models, self-supervised learning, multimodal foundation models, and model evaluation.

Amirkabir University of Technology (Tehran Polytechnic) Sep 2020 – Feb 2025
B.Sc. in Computer Science Tehran, Iran

- National University Entrance Examination Rank: 343; background in cryptography, algorithms, probability, linear algebra, AI, and software development.
- Cryptography I (18.75/20); Special Topics in Cryptography (18.25/20, highest grade in class).

Research Experience

Research Assistant – Agentic AI Security and Privacy Sep 2025 – Present
Sharif University of Technology; Supervisor: Dr. Morteza Amini Tehran, Iran

- Investigating security and privacy of LLM-based agentic and multi-agent systems, including information-flow control, authorization, provenance, memory/state, tool/skill security, and runtime enforcement.
- Studying trust boundaries, least privilege, secure composition, update safety, and recovery from unsafe effects through systems security, program analysis, formal methods, and privacy.
- Previous research covered blockchain security, cross-chain interoperability, Layer-2 systems, state verification, and protocol safety/liveness.

Teaching Experience

Teaching Assistant, Computer Networks — Sharif University of Technology; Dr. Sadeghzadeh Feb – Jun 2026

Teaching Assistant, Machine Learning — Sharif University of Technology; Dr. Motahari Feb – Jun 2026

Teaching Assistant, Artificial Intelligence and Lab — Amirkabir University of Technology; Dr. Ghatee Feb – Jun 2025

Teaching Assistant, Foundations of Matrices and Linear Algebra — Amirkabir University of Technology; Dr. Najafi Sep 2023 – Jan 2024

Teaching Assistant, Advanced Programming — Amirkabir University of Technology; Dr. Bejani Feb – Jun 2023

Technical Research Toolkit

Agentic AI & LLM Systems	LLM agents, multi-agent systems, tool/MCP/skill integration, RAG, fine-tuning, prompting and inference, agent memory/state, and agent evaluation.
Security & Formal Methods	Threat modeling, information-flow control, authorization and least privilege, provenance, taint/data-flow analysis, security invariants, runtime enforcement, SMT/model checking, and formal verification concepts.
Deep Learning	PyTorch, Transformers and attention, LLMs, state-space models, generative models, self-supervised learning, multimodal models, and model evaluation.
Cryptography & Distributed Systems	Applied cryptography, commitments, Merkle structures, zero-knowledge proof concepts, consensus, blockchain protocols, Layer-2 systems, and cross-chain architectures.
Engineering	Python, Rust, Solidity, C++, PostgreSQL, Linux, Git